THE RESTRICTED FORMAL POLYNOMIAL ALGEBRA IS REGULAR OVER A RIG-CLOSED QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof routes regularity through the excellence of the complete local quotient and its polynomial ring, applies the G-ring-to-completion regularity lemma, and composes with the smooth polynomial extension

Editorial note. The label lemma-fibre-regular names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- complete local Noetherian rings are excellent (More on Algebra, Proposition more-algebra-proposition-ubiquity-excellent) and excellence is preserved by polynomial rings
- the map from a G-ring to its completion is regular (More on Algebra, Lemma more-algebra-lemma-map-G-ring-to-completion-regular)
- smooth ring maps are regular and regular ring maps are closed under composition (More on Algebra, Section more-algebra-section-regular)

Source reference. Original lemma 0AH2: sources/stacks-chow/tags.txt, source lines 13215–13215. Prerequisite pointer invoked at proof line(s) 3630; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07QW: sources/stacks-chow/tags.txt, source lines 9846–9846. Prerequisite pointer invoked at proof line(s) 3624; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07BY: sources/stacks-chow/tags.txt, source lines 9358–9358. Prerequisite pointer invoked at proof line(s) 3621; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be an adic Noetherian topological ring. Let $\mathfrak{p} \subset A$ be a rig-closed prime. For any $n \geq 1$ the ring map*

$$A/\mathfrak{p} \longrightarrow A\{x_1, \ldots, x_n\} \otimes_A A/\mathfrak{p} = A/\mathfrak{p}\{x_1, \ldots, x_n\}$$

is regular. In particular, the algebra $A\{x_1, \ldots, x_n\} \otimes_A \kappa(\mathfrak{p})$ is geometrically regular over $\kappa(\mathfrak{p})$.

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Proof. We will use some fact on regular ring maps the reader can find in More on Algebra, Section 07BY. Since $A/\mathfrak{p}$ is a complete local Noetherian ring it is excellent (More on Algebra, Proposition 07QW). Hence $A/\mathfrak{p}[x_1, \ldots, x_n]$ is excellent (by the same reference). Hence $A/\mathfrak{p}[x_1, \ldots, x_n] \to A/\mathfrak{p}\{x_1, \ldots, x_n\}$ is a regular ring homomorphism by More on Algebra, Lemma 0AH2. Of course $A/\mathfrak{p} \to A/\mathfrak{p}[x_1, \ldots, x_n]$ is smooth and hence regular. Since the composition of regular ring maps is regular the proof is complete. $\square$